On Numerosity of Deep Neural Networks

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NEURAL INFORMATION PROCESSING SYSTEMS

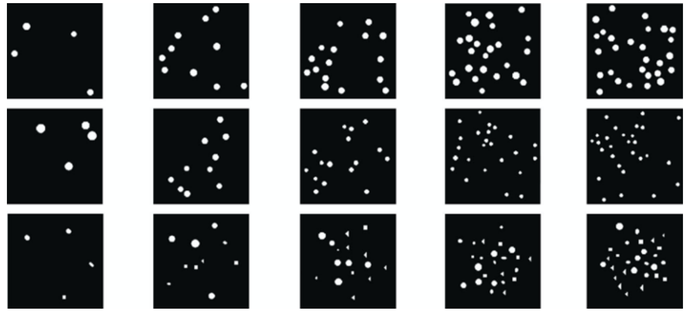
Key Result

Recently, a provocative claim was published that number sense spontaneously emerges in a deep neural network trained merely for visual object recognition [Nasr *et al.* 2019]. **But the above claim is incorrect.**

- The sample set used to discover number-aware neurons is too small, compared to the huge number of neurons in the object recognition network.
- This erroneous approach could find number-sensing neurons even in randomly initialized deep neural networks that are not trained at all.
- Even after being trained with number-depicting images, deep learning still has difficulties to acquire the abstract concept of numbers.

Review of Nasr *et al.*'s Work

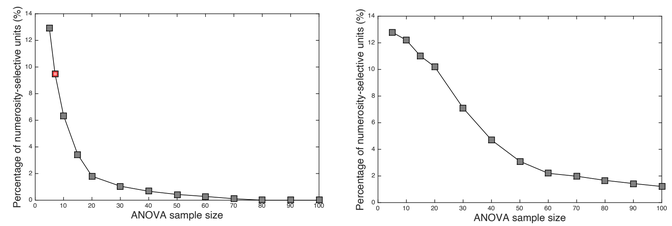
- Train a DCNN for visual object recognition using the ImageNet dataset.
- Feed the test images that are binary visual representations of numbers to the network and record the responses of the final convolutional layer.
- Perform a two-way analysis of variance (ANOVA) to detect numerosity-selective network units.
- Of the 37632 network units in the final convolutional layer, 3601 (9.6%) are found to be numerosity-selective.
- The responses of these numerosity-selective units exhibit a clear tuning pattern that is virtually identical to those of real neurons from monkey prefrontal cortex.



Sample images in the test set. Top: Standard set contains images displaying objects as circular dots of random size and spacing. Middle: Control set 1 contains images displaying dots of equal total area and the same dot density. Bottom: Control set 2 contains images displaying objects of different geometric shapes with an equal area of the convex hull.

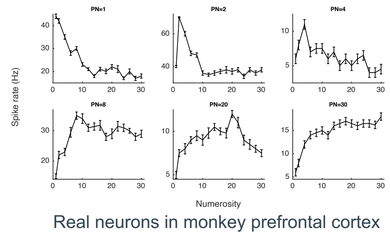
Critiques of General DCNN Numerosity

- The number of numerosity-selective neurons decreases drastically in the sample size of ANOVA .

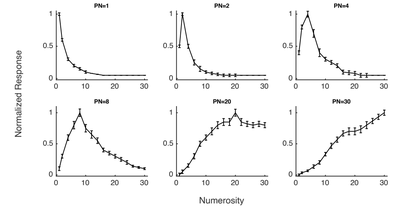


Left: Network trained for visual object recognition. Right: Untrained network with initialized weights from a uniform distribution U(−0.1, 0.1).

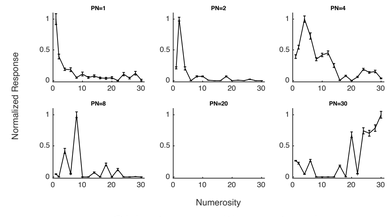
- The numerosity-selective neurons exist even in a randomly initialized network that is not trained for any task at all, if this hypothesis is tested with a small enough number of numerosity images.



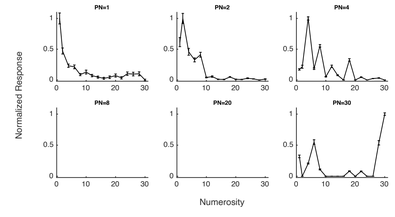
Real neurons in monkey prefrontal cortex



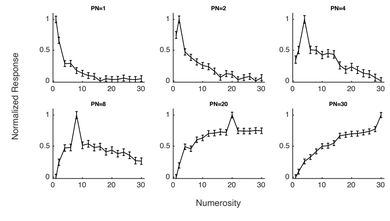
Object recognition network
ANOVA sample size = 7



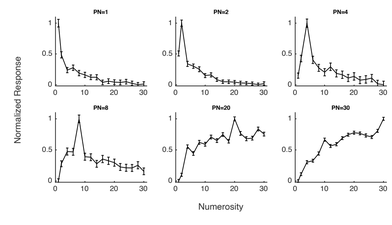
Object recognition network
ANOVA sample size = 30



Object recognition network
ANOVA sample size = 50



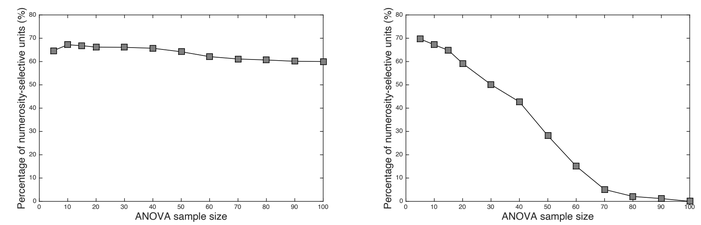
Initialized weights from U(−0.1, 0.1)
ANOVA sample size = 7



Initialized weights from N(0, 0.05)
ANOVA sample size = 7

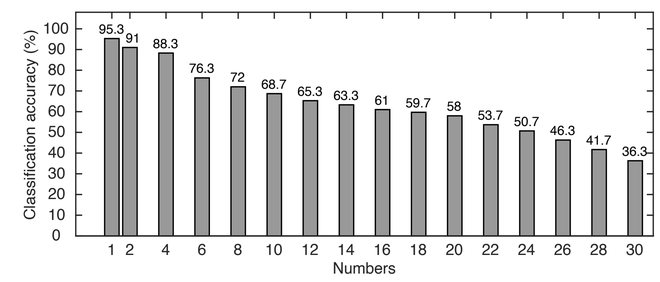
Poor Generalization of Neural Networks Trained for Numerosity (Nu-Net)

- The number of numerosity-selective neurons decreases steadily when analyzed with the superficially varied test images.



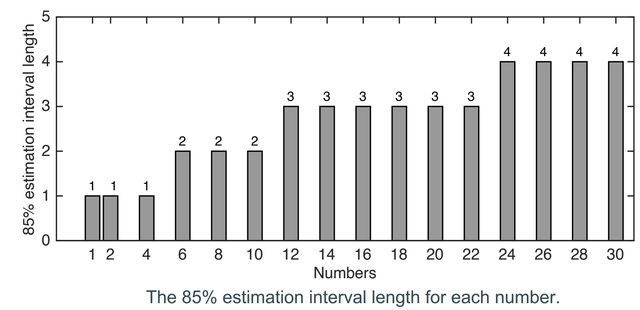
Left: Test images are drawn from the same distribution of the training images.
Right: Test images have 50% greater variations in object size than training images.

- The above modestly increased variability in object size decreases the average classification accuracy of the Nu-Net.



Classification accuracy of the Nu-Net for each number on the test images that have 50% greater variations in object size than the training images.

- The Nu-Net performs well on small numbers (1, 2 and 4). Starting from 6, the 85% estimation interval length increases, indicating an increasing level of uncertainty in number perception for large populations.



The 85% estimation interval length for each number.

References

Khaled Nasr et al. (2019). "Number detectors spontaneously emerge in a deep neural network designed for visual object recognition." In: Science Advances. Vol. 5, no. 5, eaav7903.